IET, C-DAC

500 MCQ PRACTICE QUESTION BANK

Database Technologies

Moderate to Difficult | Conceptual | Code Output | Error Analysis | Tricky Applications

Purpose

A subject-wise practice resource for engineering graduates preparing for C-DAC course examinations. Questions are freshly authored for practice and are not represented as actual or leaked C-DAC examination questions.

Coverage

Relational model and ER design; keys and constraints; normalization; SQL DDL/DML/DCL/TCL; joins and subqueries; aggregates and window functions; indexes and query plans; transactions and isolation; procedures and triggers; NoSQL and MongoDB.

Instructions

Each question has four options and one best answer. Attempt the questions before consulting the consolidated answer key at the end. For code questions, assume standard language behavior and the stated code fragment is complete for the operation being tested.

Question count: 500 **Options:** A–D **Answer key:** Included

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QUESTION PAPER

Database Technologies — 500 Questions

Q1.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. Only matching rows from A and B are retained
C. Every row from A is retained; unmatched B columns are NULL
D. The result is always a Cartesian product

Q2.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Consistency
C. Durability
D. Isolation

Q3.

For values [87, 78, 71], what is COUNT(*) returned by WHERE value > 78?

A. 2
B. 2
C. 1
D. 3

Q4.

A table contains three non-NULL values [13, 57, 53]. What is AVG(amount) for these rows?

A. 123
B. 41.00
C. 44.33
D. 57

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Q5.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE city = 'Pune'
C. WHERE last_name = 'Patel'
D. WHERE first_name = 'Asha' only

Q6.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A partial dependency on part of the candidate key
B. A multivalued dependency only
C. A transitive dependency through a non-key attribute
D. No dependency is violated

Q7.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A multivalued dependency only
C. A transitive dependency through a non-key attribute
D. A partial dependency on part of the candidate key

Q8.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE last_name = 'Patel'
B. WHERE city = 'Pune'
C. WHERE salary = 50000
D. WHERE first_name = 'Asha' only

Q9.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE salary = 50000
C. WHERE last_name = 'Patel'
D. WHERE first_name = 'Asha' only

Q10.

What is the purpose of a foreign key?

A. Encrypt a column
B. Guarantee that every column is unique
C. Enforce referential integrity between related tables
D. Sort rows physically

Q11.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. Only matching rows from A and B are retained
C. The result is always a Cartesian product
D. Every row from A is retained; unmatched B columns are NULL

Q12.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission <> NULL
C. commission == NULL
D. commission = NULL

Q13.

Consider salaries [49, 12, 78]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 90
B. 78
C. 127
D. 139

Q14.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A multivalued dependency only
C. No dependency is violated
D. A partial dependency on part of the candidate key

Q15.

For values [69, 79, 56], what is COUNT(*) returned by WHERE value > 79?

A. 3
B. 1
C. 3
D. 0

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Q16.

Consider salaries [61, 72, 78]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 283
B. 211
C. 272
D. 211

Q17.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission == NULL
C. commission <> NULL
D. commission = NULL

Q18.

Consider salaries [75, 32, 22]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 107
B. 150
C. 129
D. 75

Q19.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Isolation
C. Atomicity
D. Consistency

Q20.

For values [82, 72, 24], what is COUNT(*) returned by WHERE value > 72?

A. 2
B. 3
C. 1
D. 2

Q21.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from A is retained; unmatched B columns are NULL
C. Every row from B is retained
D. Only matching rows from A and B are retained

Q22.

A table contains three non-NULL values [24, 18, 67]. What is AVG(amount) for these rows?

A. 36.33
B. 39.67
C. 109
D. 67

Q23.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Isolation
B. Atomicity
C. Durability
D. Consistency

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Q24.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission == NULL
B. commission IS NULL
C. commission = NULL
D. commission <> NULL

Q25.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission = NULL
C. commission <> NULL
D. commission == NULL

Q26.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from A is retained; unmatched B columns are NULL
C. Only matching rows from A and B are retained
D. Every row from B is retained

Q27.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. A transitive dependency through a non-key attribute
C. No dependency is violated
D. A partial dependency on part of the candidate key

Q28.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission == NULL
C. commission IS NULL
D. commission = NULL

Q29.

For values [68, 81, 80], what is COUNT(*) returned by WHERE value > 81?

A. 3
B. 1
C. 0
D. 3

Q30.

For values [84, 68, 16], what is COUNT(*) returned by WHERE value > 68?

A. 3
B. 2
C. 1
D. 2

Q31.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Durability
C. Consistency
D. Isolation

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Q32.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. Every row from A is retained; unmatched B columns are NULL
C. The result is always a Cartesian product
D. Only matching rows from A and B are retained

Q33.

For values [24, 66, 34], what is COUNT(*) returned by WHERE value > 66?

A. 3
B. 0
C. 1
D. 3

Q34.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A partial dependency on part of the candidate key
B. A multivalued dependency only
C. A transitive dependency through a non-key attribute
D. No dependency is violated

Q35.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE last_name = 'Patel'
B. WHERE salary = 50000
C. WHERE city = 'Pune'
D. WHERE first_name = 'Asha' only

Q36.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission == NULL
B. commission <> NULL
C. commission IS NULL
D. commission = NULL

Q37.

For values [42, 10, 40], what is COUNT(*) returned by WHERE value > 10?

A. 3
B. 1
C. 2
D. 3

Q38.

A table contains three non-NULL values [10, 58, 54]. What is AVG(amount) for these rows?

A. 122
B. 40.67
C. 44.00
D. 58

Q39.

For values [56, 53, 56], what is COUNT(*) returned by WHERE value > 53?

A. 2
B. 3
C. 1
D. 3

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Q40.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. The result is always a Cartesian product
C. Every row from A is retained; unmatched B columns are NULL
D. Only matching rows from A and B are retained

Q41.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission == NULL
C. commission = NULL
D. commission IS NULL

Q42.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE salary = 50000
C. WHERE first_name = 'Asha' only
D. WHERE last_name = 'Patel'

Q43.

A table contains three non-NULL values [84, 90, 71]. What is AVG(amount) for these rows?

A. 81.67
B. 90
C. 245
D. 85.00

Q44.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. The result is always a Cartesian product
C. Only matching rows from A and B are retained
D. Every row from A is retained; unmatched B columns are NULL

Q45.

Which index is most likely to help a query filtering on a selective column?

A. An index whose leading key matches the predicate
B. An index on an unrelated column
C. An index on every table's row count
D. No index can help filters

Q46.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from A is retained; unmatched B columns are NULL
B. Every row from B is retained
C. Only matching rows from A and B are retained
D. The result is always a Cartesian product

Q47.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A partial dependency on part of the candidate key
C. A multivalued dependency only
D. No dependency is violated

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Q48.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE last_name = 'Patel'
B. WHERE salary = 50000
C. WHERE first_name = 'Asha' only
D. WHERE city = 'Pune'

Q49.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission == NULL
C. commission IS NULL
D. commission = NULL

Q50.

Consider salaries [15, 19, 72]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 91
B. 106
C. 87
D. 72

Q51.

A table contains three non-NULL values [22, 30, 14]. What is AVG(amount) for these rows?

A. 66
B. 30
C. 22.00
D. 25.33

Q52.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Consistency
C. Isolation
D. Durability

Q53.

Consider salaries [76, 32, 48]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 156
B. 152
C. 108
D. 76

Q54.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE city = 'Pune'
C. WHERE last_name = 'Patel'
D. WHERE salary = 50000

Q55.

Consider salaries [23, 63, 19]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 105
B. 126
C. 63
D. 86

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Q56.

What is a dirty read?

A. Reading from a backup
B. Reading a NULL value
C. Reading a row twice after commit
D. Reading data written by an uncommitted transaction

Q57.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from A is retained; unmatched B columns are NULL
B. Every row from B is retained
C. Only matching rows from A and B are retained
D. The result is always a Cartesian product

Q58.

A table contains three non-NULL values [59, 78, 50]. What is AVG(amount) for these rows?

A. 78
B. 187
C. 62.33
D. 65.67

Q59.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission = NULL
C. commission IS NULL
D. commission == NULL

Q60.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Durability
C. Isolation
D. Consistency

Q61.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE last_name = 'Patel'
B. WHERE first_name = 'Asha' only
C. WHERE city = 'Pune'
D. WHERE salary = 50000

Q62.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A multivalued dependency only
C. No dependency is violated
D. A partial dependency on part of the candidate key

Q63.

Consider salaries [79, 10, 46]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 135
B. 79
C. 89
D. 158

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Q64.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A partial dependency on part of the candidate key
C. A multivalued dependency only
D. No dependency is violated

Q65.

A table contains three non-NULL values [25, 34, 86]. What is AVG(amount) for these rows?

A. 145
B. 51.67
C. 86
D. 48.33

Q66.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from A is retained; unmatched B columns are NULL
C. Only matching rows from A and B are retained
D. Every row from B is retained

Q67.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A multivalued dependency only
C. A partial dependency on part of the candidate key
D. A transitive dependency through a non-key attribute

Q68.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A partial dependency on part of the candidate key
B. No dependency is violated
C. A transitive dependency through a non-key attribute
D. A multivalued dependency only

Q69.

Which MongoDB structure stores documents?

A. Sequence
B. Schema-only table
C. Cursor index file
D. Collection

Q70.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A multivalued dependency only
C. A transitive dependency through a non-key attribute
D. A partial dependency on part of the candidate key

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Q71.

A table contains three non-NULL values [45, 78, 64]. What is AVG(amount) for these rows?

A. 65.67
B. 187
C. 62.33
D. 78

Q72.

What is the main purpose of a database trigger?

A. Create a client-side cache
B. Automatically execute logic in response to specified database events
C. Schedule operating-system tasks
D. Replace all constraints

Q73.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE last_name = 'Patel'
B. WHERE salary = 50000
C. WHERE city = 'Pune'
D. WHERE first_name = 'Asha' only

Q74.

Consider salaries [67, 51, 28]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 146
B. 118
C. 185
D. 169

Q75.

Which command makes a transaction's changes permanent?

A. GRANT
B. SAVEPOINT
C. COMMIT
D. ROLLBACK

Q76.

Consider salaries [73, 21, 89]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 235
B. 162
C. 183
D. 183

Q77.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. Only matching rows from A and B are retained
C. Every row from A is retained; unmatched B columns are NULL
D. The result is always a Cartesian product

Q78.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from A is retained; unmatched B columns are NULL
C. Only matching rows from A and B are retained
D. Every row from B is retained

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Q79.

A table contains three non-NULL values [30, 24, 51]. What is AVG(amount) for these rows?

A. 38.33
B. 105
C. 35.00
D. 51

Q80.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission IS NULL
C. commission == NULL
D. commission = NULL

Q81.

A table contains three non-NULL values [53, 75, 19]. What is AVG(amount) for these rows?

A. 49.00
B. 147
C. 52.33
D. 75

Q82.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE last_name = 'Patel'
C. WHERE first_name = 'Asha' only
D. WHERE salary = 50000

Q83.

A table contains three non-NULL values [67, 77, 64]. What is AVG(amount) for these rows?

A. 72.67
B. 69.33
C. 208
D. 77

Q84.

For values [55, 24, 40], what is COUNT(*) returned by WHERE value > 24?

A. 1
B. 3
C. 2
D. 3

Q85.

Consider salaries [39, 10, 36]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 85
B. 39
C. 0
D. 10

Q86.

A table contains three non-NULL values [63, 74, 33]. What is AVG(amount) for these rows?

A. 74
B. 170
C. 56.67
D. 60.00

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Q87.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A multivalued dependency only
C. A partial dependency on part of the candidate key
D. A transitive dependency through a non-key attribute

Q88.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A partial dependency on part of the candidate key
B. No dependency is violated
C. A multivalued dependency only
D. A transitive dependency through a non-key attribute

Q89.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A partial dependency on part of the candidate key
C. A transitive dependency through a non-key attribute
D. A multivalued dependency only

Q90.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission == NULL
B. commission = NULL
C. commission <> NULL
D. commission IS NULL

Q91.

Consider salaries [26, 85, 69]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 239
B. 180
C. 154
D. 180

Q92.

Consider salaries [86, 75, 62]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 309
B. 223
C. 298
D. 223

Q93.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission == NULL
C. commission IS NULL
D. commission <> NULL

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Q94.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE first_name = 'Asha' only
C. WHERE last_name = 'Patel'
D. WHERE salary = 50000

Q95.

Consider salaries [60, 77, 64]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 201
B. 201
C. 278
D. 261

Q96.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission IS NULL
C. commission == NULL
D. commission = NULL

Q97.

For values [10, 57, 49], what is COUNT(*) returned by WHERE value > 57?

A. 0
B. 3
C. 1
D. 3

Q98.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. The result is always a Cartesian product
C. Every row from B is retained
D. Every row from A is retained; unmatched B columns are NULL

Q99.

A table contains three non-NULL values [14, 73, 38]. What is AVG(amount) for these rows?

A. 73
B. 125
C. 45.00
D. 41.67

Q100.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Isolation
B. Durability
C. Consistency
D. Atomicity

Q101.

What does COUNT(column_name) do with NULL values?

A. It counts distinct values only
B. It counts every row
C. It ignores NULL values
D. It returns NULL if any value is NULL

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Q102.

What is a non-repeatable read?

A. An index is rebuilt
B. A transaction reads uncommitted data
C. A row read twice has different committed values due to another transaction
D. A query returns duplicate column names

Q103.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from B is retained
C. Every row from A is retained; unmatched B columns are NULL
D. Only matching rows from A and B are retained

Q104.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A partial dependency on part of the candidate key
B. No dependency is violated
C. A multivalued dependency only
D. A transitive dependency through a non-key attribute

Q105.

Consider salaries [87, 39, 78]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 204
B. 204
C. 165
D. 252

Q106.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from B is retained
C. Only matching rows from A and B are retained
D. Every row from A is retained; unmatched B columns are NULL

Q107.

A table contains three non-NULL values [61, 67, 50]. What is AVG(amount) for these rows?

A. 59.33
B. 178
C. 67
D. 62.67

Q108.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission = NULL
C. commission <> NULL
D. commission == NULL

Q109.

For values [65, 24, 31], what is COUNT(*) returned by WHERE value > 24?

A. 3
B. 3
C. 2
D. 1

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Q110.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission = NULL
C. commission IS NULL
D. commission == NULL

Q111.

A table contains three non-NULL values [56, 30, 59]. What is AVG(amount) for these rows?

A. 48.33
B. 59
C. 51.67
D. 145

Q112.

A table contains three non-NULL values [47, 85, 39]. What is AVG(amount) for these rows?

A. 60.33
B. 171
C. 85
D. 57.00

Q113.

A table contains three non-NULL values [90, 82, 33]. What is AVG(amount) for these rows?

A. 90
B. 205
C. 71.67
D. 68.33

Q114.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. Only matching rows from A and B are retained
C. Every row from A is retained; unmatched B columns are NULL
D. The result is always a Cartesian product

Q115.

For values [65, 77, 12], what is COUNT(*) returned by WHERE value > 77?

A. 1
B. 3
C. 0
D. 3

Q116.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. Every row from A is retained; unmatched B columns are NULL
C. The result is always a Cartesian product
D. Only matching rows from A and B are retained

Q117.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Isolation
B. Durability
C. Atomicity
D. Consistency

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Q118.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Isolation
C. Consistency
D. Durability

Q119.

A table contains three non-NULL values [45, 50, 71]. What is AVG(amount) for these rows?

A. 71
B. 166
C. 55.33
D. 58.67

Q120.

For values [46, 30, 41], what is COUNT(*) returned by WHERE value > 30?

A. 1
B. 2
C. 3
D. 3

Q121.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A partial dependency on part of the candidate key
C. A multivalued dependency only
D. No dependency is violated

Q122.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from A is retained; unmatched B columns are NULL
B. The result is always a Cartesian product
C. Every row from B is retained
D. Only matching rows from A and B are retained

Q123.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Durability
C. Isolation
D. Atomicity

Q124.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE city = 'Pune'
C. WHERE salary = 50000
D. WHERE last_name = 'Patel'

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Q125.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Isolation
C. Atomicity
D. Durability

Q126.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission IS NULL
C. commission == NULL
D. commission <> NULL

Q127.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A partial dependency on part of the candidate key
B. No dependency is violated
C. A multivalued dependency only
D. A transitive dependency through a non-key attribute

Q128.

Which clause limits rows before grouping and aggregation?

A. HAVING
B. WINDOW
C. ORDER BY
D. WHERE

Q129.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from A is retained; unmatched B columns are NULL
B. Only matching rows from A and B are retained
C. The result is always a Cartesian product
D. Every row from B is retained

Q130.

Which SQL function returns the current date in many SQL dialects?

A. GROUP_CONCAT
B. CURRENT_DATE
C. COALESCE
D. COUNT(*)

Q131.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission <> NULL
C. commission == NULL
D. commission = NULL

Q132.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE first_name = 'Asha' only
C. WHERE last_name = 'Patel'
D. WHERE city = 'Pune'

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Q133.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission <> NULL
C. commission == NULL
D. commission = NULL

Q134.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE salary = 50000
C. WHERE first_name = 'Asha' only
D. WHERE last_name = 'Patel'

Q135.

A table contains three non-NULL values [52, 43, 66]. What is AVG(amount) for these rows?

A. 57.00
B. 66
C. 161
D. 53.67

Q136.

Consider salaries [67, 22, 30]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 119
B. 67
C. 89
D. 134

Q137.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A partial dependency on part of the candidate key
C. A multivalued dependency only
D. No dependency is violated

Q138.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission == NULL
B. commission = NULL
C. commission IS NULL
D. commission <> NULL

Q139.

For values [72, 41, 59], what is COUNT(*) returned by WHERE value > 41?

A. 3
B. 3
C. 1
D. 2

Q140.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Isolation
B. Durability
C. Consistency
D. Atomicity

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Q141.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. Every row from B is retained
C. The result is always a Cartesian product
D. Every row from A is retained; unmatched B columns are NULL

Q142.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE first_name = 'Asha' only
C. WHERE city = 'Pune'
D. WHERE last_name = 'Patel'

Q143.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE last_name = 'Patel'
C. WHERE city = 'Pune'
D. WHERE salary = 50000

Q144.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A partial dependency on part of the candidate key
C. No dependency is violated
D. A multivalued dependency only

Q145.

A table contains three non-NULL values [77, 37, 21]. What is AVG(amount) for these rows?

A. 45.00
B. 48.33
C. 77
D. 135

Q146.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Isolation
C. Atomicity
D. Durability

Q147.

A table contains three non-NULL values [33, 56, 22]. What is AVG(amount) for these rows?

A. 37.00
B. 56
C. 40.33
D. 111

Q148.

Consider salaries [32, 12, 69]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 113
B. 69
C. 81
D. 101

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Q149.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission <> NULL
C. commission == NULL
D. commission = NULL

Q150.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Consistency
C. Atomicity
D. Isolation

Q151.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Consistency
C. Isolation
D. Durability

Q152.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE last_name = 'Patel'
C. WHERE first_name = 'Asha' only
D. WHERE salary = 50000

Q153.

Which SQL expression correctly tests for missing values?

A. NULL()
B. IS NULL
C. = NULL
D. == NULL

Q154.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from A is retained; unmatched B columns are NULL
C. Every row from B is retained
D. Only matching rows from A and B are retained

Q155.

What is a candidate key?

A. A non-unique index
B. A foreign key that references itself
C. A minimal set of attributes that uniquely identifies a row
D. Any nullable column

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Q156.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A partial dependency on part of the candidate key
B. A multivalued dependency only
C. No dependency is violated
D. A transitive dependency through a non-key attribute

Q157.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. The result is always a Cartesian product
C. Every row from A is retained; unmatched B columns are NULL
D. Every row from B is retained

Q158.

Which index structure is commonly used by relational databases for ordered range lookups?

A. Stack
B. Bloom filter only
C. B-tree/B+ tree
D. Trie of SQL keywords

Q159.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. No dependency is violated
C. A multivalued dependency only
D. A partial dependency on part of the candidate key

Q160.

Consider salaries [74, 89, 40]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 252
B. 163
C. 237
D. 203

Q161.

A table contains three non-NULL values [23, 15, 62]. What is AVG(amount) for these rows?

A. 100
B. 62
C. 36.67
D. 33.33

Q162.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE last_name = 'Patel'
C. WHERE salary = 50000
D. WHERE city = 'Pune'

Q163.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Only matching rows from A and B are retained
C. Every row from B is retained
D. Every row from A is retained; unmatched B columns are NULL

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Q164.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. A transitive dependency through a non-key attribute
C. No dependency is violated
D. A partial dependency on part of the candidate key

Q165.

Which normal form removes partial dependencies on a candidate key?

A. Second Normal Form (2NF)
B. BCNF only
C. First Normal Form (1NF)
D. Third Normal Form (3NF)

Q166.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from A is retained; unmatched B columns are NULL
B. Every row from B is retained
C. The result is always a Cartesian product
D. Only matching rows from A and B are retained

Q167.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A multivalued dependency only
C. A partial dependency on part of the candidate key
D. No dependency is violated

Q168.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A partial dependency on part of the candidate key
B. A transitive dependency through a non-key attribute
C. A multivalued dependency only
D. No dependency is violated

Q169.

A table contains three non-NULL values [85, 66, 15]. What is AVG(amount) for these rows?

A. 166
B. 55.33
C. 58.67
D. 85

Q170.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Atomicity
C. Isolation
D. Durability

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Q171.

Consider salaries [73, 21, 22]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 94
B. 116
C. 73
D. 146

Q172.

A table contains three non-NULL values [75, 81, 30]. What is AVG(amount) for these rows?

A. 62.00
B. 65.33
C. 81
D. 186

Q173.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE salary = 50000
C. WHERE city = 'Pune'
D. WHERE last_name = 'Patel'

Q174.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission == NULL
C. commission IS NULL
D. commission <> NULL

Q175.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission == NULL
C. commission = NULL
D. commission <> NULL

Q176.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission == NULL
B. commission <> NULL
C. commission IS NULL
D. commission = NULL

Q177.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Isolation
C. Atomicity
D. Consistency

Q178.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE salary = 50000
C. WHERE last_name = 'Patel'
D. WHERE city = 'Pune'

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Q179.

A table contains three non-NULL values [49, 24, 22]. What is AVG(amount) for these rows?

A. 35.00
B. 49
C. 31.67
D. 95

Q180.

Which normal form requires every determinant to be a candidate key?

A. 1NF
B. BCNF
C. 2NF
D. 4NF

Q181.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE city = 'Pune'
C. WHERE first_name = 'Asha' only
D. WHERE last_name = 'Patel'

Q182.

For values [46, 67, 23], what is COUNT(*) returned by WHERE value > 67?

A. 3
B. 1
C. 3
D. 0

Q183.

Consider salaries [85, 41, 69]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 195
B. 154
C. 239
D. 195

Q184.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE city = 'Pune'
C. WHERE last_name = 'Patel'
D. WHERE salary = 50000

Q185.

A table contains three non-NULL values [84, 47, 34]. What is AVG(amount) for these rows?

A. 84
B. 58.33
C. 165
D. 55.00

Q186.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE salary = 50000
C. WHERE last_name = 'Patel'
D. WHERE city = 'Pune'

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Q187.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE last_name = 'Patel'
C. WHERE first_name = 'Asha' only
D. WHERE city = 'Pune'

Q188.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A partial dependency on part of the candidate key
C. A multivalued dependency only
D. No dependency is violated

Q189.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. No dependency is violated
C. A transitive dependency through a non-key attribute
D. A partial dependency on part of the candidate key

Q190.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. A partial dependency on part of the candidate key
C. No dependency is violated
D. A transitive dependency through a non-key attribute

Q191.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Consistency
C. Atomicity
D. Isolation

Q192.

Consider salaries [55, 64, 47]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 166
B. 119
C. 183
D. 174

Q193.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from A is retained; unmatched B columns are NULL
B. Only matching rows from A and B are retained
C. The result is always a Cartesian product
D. Every row from B is retained

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Q194.

What does ACID atomicity mean?

A. Every query uses an index
B. Data is stored in alphabetical order
C. A transaction is all-or-nothing
D. Transactions always run in parallel

Q195.

A table contains three non-NULL values [86, 36, 17]. What is AVG(amount) for these rows?

A. 49.67
B. 86
C. 139
D. 46.33

Q196.

For values [45, 63, 88], what is COUNT(*) returned by WHERE value > 63?

A. 2
B. 3
C. 1
D. 2

Q197.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Isolation
C. Consistency
D. Durability

Q198.

A table contains three non-NULL values [29, 48, 45]. What is AVG(amount) for these rows?

A. 44.00
B. 48
C. 122
D. 40.67

Q199.

For values [43, 40, 64], what is COUNT(*) returned by WHERE value > 40?

A. 3
B. 1
C. 3
D. 2

Q200.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Isolation
C. Atomicity
D. Durability

Q201.

For values [30, 86, 27], what is COUNT(*) returned by WHERE value > 86?

A. 1
B. 0
C. 3
D. 3

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Q202.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Only matching rows from A and B are retained
C. Every row from B is retained
D. Every row from A is retained; unmatched B columns are NULL

Q203.

Consider salaries [31, 82, 38]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 113
B. 164
C. 82
D. 151

Q204.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Only matching rows from A and B are retained
C. Every row from A is retained; unmatched B columns are NULL
D. Every row from B is retained

Q205.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from A is retained; unmatched B columns are NULL
B. The result is always a Cartesian product
C. Only matching rows from A and B are retained
D. Every row from B is retained

Q206.

Consider salaries [22, 52, 47]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 121
B. 52
C. 74
D. 104

Q207.

Consider salaries [65, 30, 16]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 65
B. 95
C. 111
D. 130

Q208.

A table contains three non-NULL values [47, 34, 78]. What is AVG(amount) for these rows?

A. 53.00
B. 78
C. 159
D. 56.33

Q209.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. A partial dependency on part of the candidate key
C. No dependency is violated
D. A transitive dependency through a non-key attribute

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Q210.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. Only matching rows from A and B are retained
C. Every row from A is retained; unmatched B columns are NULL
D. The result is always a Cartesian product

Q211.

For values [60, 62, 67], what is COUNT(*) returned by WHERE value > 62?

A. 1
B. 3
C. 2
D. 2

Q212.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission == NULL
B. commission IS NULL
C. commission <> NULL
D. commission = NULL

Q213.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission <> NULL
C. commission = NULL
D. commission == NULL

Q214.

A table contains three non-NULL values [76, 49, 29]. What is AVG(amount) for these rows?

A. 154
B. 76
C. 54.67
D. 51.33

Q215.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. The result is always a Cartesian product
C. Every row from A is retained; unmatched B columns are NULL
D. Only matching rows from A and B are retained

Q216.

For values [11, 71, 69], what is COUNT(*) returned by WHERE value > 71?

A. 3
B. 0
C. 3
D. 1

Q217.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. No dependency is violated
C. A partial dependency on part of the candidate key
D. A transitive dependency through a non-key attribute

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Q218.

What is denormalization?

A. Intentionally introducing redundancy to improve selected read workloads
B. Removing all keys
C. Converting SQL into JSON
D. Splitting every table into one-column tables

Q219.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Durability
C. Atomicity
D. Isolation

Q220.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE last_name = 'Patel'
B. WHERE first_name = 'Asha' only
C. WHERE salary = 50000
D. WHERE city = 'Pune'

Q221.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from A is retained; unmatched B columns are NULL
C. Only matching rows from A and B are retained
D. Every row from B is retained

Q222.

For values [81, 35, 76], what is COUNT(*) returned by WHERE value > 35?

A. 3
B. 1
C. 3
D. 2

Q223.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission <> NULL
C. commission IS NULL
D. commission == NULL

Q224.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. A partial dependency on part of the candidate key
C. No dependency is violated
D. A transitive dependency through a non-key attribute

Q225.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from A is retained; unmatched B columns are NULL
C. Only matching rows from A and B are retained
D. Every row from B is retained

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Q226.

For values [32, 89, 55], what is COUNT(*) returned by WHERE value > 89?

A. 3
B. 0
C. 1
D. 3

Q227.

A table contains three non-NULL values [21, 58, 61]. What is AVG(amount) for these rows?

A. 140
B. 61
C. 46.67
D. 50.00

Q228.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission == NULL
B. commission IS NULL
C. commission = NULL
D. commission <> NULL

Q229.

Which normal form addresses transitive dependencies of non-key attributes on a key?

A. 2NF
B. 1NF
C. Third Normal Form (3NF)
D. 0NF

Q230.

For values [81, 50, 88], what is COUNT(*) returned by WHERE value > 50?

A. 1
B. 3
C. 2
D. 3

Q231.

For values [88, 24, 86], what is COUNT(*) returned by WHERE value > 24?

A. 1
B. 3
C. 3
D. 2

Q232.

Consider salaries [80, 76, 80]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 316
B. 236
C. 312
D. 236

Q233.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Consistency
C. Durability
D. Isolation

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Q234.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE salary = 50000
C. WHERE first_name = 'Asha' only
D. WHERE last_name = 'Patel'

Q235.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE salary = 50000
C. WHERE last_name = 'Patel'
D. WHERE first_name = 'Asha' only

Q236.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission IS NULL
C. commission = NULL
D. commission == NULL

Q237.

For values [79, 73, 43], what is COUNT(*) returned by WHERE value > 73?

A. 2
B. 1
C. 3
D. 2

Q238.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Durability
C. Isolation
D. Consistency

Q239.

Consider salaries [18, 53, 25]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 106
B. 96
C. 71
D. 53

Q240.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission = NULL
C. commission <> NULL
D. commission == NULL

Q241.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. Every row from A is retained; unmatched B columns are NULL
C. Every row from B is retained
D. The result is always a Cartesian product

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Q242.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. Every row from B is retained
C. The result is always a Cartesian product
D. Every row from A is retained; unmatched B columns are NULL

Q243.

For values [37, 29, 83], what is COUNT(*) returned by WHERE value > 29?

A. 3
B. 2
C. 3
D. 1

Q244.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A partial dependency on part of the candidate key
C. A multivalued dependency only
D. No dependency is violated

Q245.

For values [59, 52, 40], what is COUNT(*) returned by WHERE value > 52?

A. 1
B. 2
C. 2
D. 3

Q246.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission == NULL
C. commission IS NULL
D. commission = NULL

Q247.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission = NULL
C. commission <> NULL
D. commission == NULL

Q248.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from A is retained; unmatched B columns are NULL
B. Only matching rows from A and B are retained
C. Every row from B is retained
D. The result is always a Cartesian product

Q249.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. A partial dependency on part of the candidate key
C. A transitive dependency through a non-key attribute
D. No dependency is violated

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Q250.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE first_name = 'Asha' only
C. WHERE last_name = 'Patel'
D. WHERE city = 'Pune'

Q251.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission == NULL
C. commission = NULL
D. commission <> NULL

Q252.

For values [27, 55, 18], what is COUNT(*) returned by WHERE value > 55?

A. 3
B. 0
C. 1
D. 3

Q253.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission IS NULL
C. commission <> NULL
D. commission == NULL

Q254.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission = NULL
C. commission IS NULL
D. commission == NULL

Q255.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE last_name = 'Patel'
C. WHERE city = 'Pune'
D. WHERE first_name = 'Asha' only

Q256.

Consider salaries [85, 28, 35]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 85
B. 148
C. 113
D. 170

Q257.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission IS NULL
C. commission <> NULL
D. commission == NULL

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Q258.

For values [48, 24, 19], what is COUNT(*) returned by WHERE value > 24?

A. 2
B. 1
C. 3
D. 2

Q259.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. Only matching rows from A and B are retained
C. The result is always a Cartesian product
D. Every row from A is retained; unmatched B columns are NULL

Q260.

For values [51, 15, 90], what is COUNT(*) returned by WHERE value > 15?

A. 1
B. 2
C. 3
D. 3

Q261.

For values [29, 68, 84], what is COUNT(*) returned by WHERE value > 68?

A. 2
B. 1
C. 3
D. 2

Q262.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission == NULL
B. commission <> NULL
C. commission IS NULL
D. commission = NULL

Q263.

For values [35, 71, 42], what is COUNT(*) returned by WHERE value > 71?

A. 0
B. 3
C. 3
D. 1

Q264.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission == NULL
C. commission = NULL
D. commission IS NULL

Q265.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Durability
C. Atomicity
D. Isolation

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Q266.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Isolation
C. Atomicity
D. Consistency

Q267.

For values [34, 10, 25], what is COUNT(*) returned by WHERE value > 10?

A. 3
B. 2
C. 1
D. 3

Q268.

A table contains three non-NULL values [22, 71, 44]. What is AVG(amount) for these rows?

A. 49.00
B. 137
C. 45.67
D. 71

Q269.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Isolation
B. Consistency
C. Atomicity
D. Durability

Q270.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from B is retained
C. Every row from A is retained; unmatched B columns are NULL
D. Only matching rows from A and B are retained

Q271.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE last_name = 'Patel'
C. WHERE first_name = 'Asha' only
D. WHERE salary = 50000

Q272.

For values [86, 35, 59], what is COUNT(*) returned by WHERE value > 35?

A. 3
B. 2
C. 3
D. 1

Q273.

For values [51, 52, 21], what is COUNT(*) returned by WHERE value > 52?

A. 1
B. 3
C. 0
D. 3

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Q274.

Consider salaries [61, 47, 47]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 61
B. 108
C. 155
D. 122

Q275.

Consider salaries [35, 86, 46]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 172
B. 121
C. 86
D. 167

Q276.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Isolation
C. Consistency
D. Durability

Q277.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission = NULL
C. commission == NULL
D. commission <> NULL

Q278.

For values [57, 57, 64], what is COUNT(*) returned by WHERE value > 57?

A. 1
B. 2
C. 3
D. 2

Q279.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. Every row from A is retained; unmatched B columns are NULL
C. Only matching rows from A and B are retained
D. The result is always a Cartesian product

Q280.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE first_name = 'Asha' only
C. WHERE last_name = 'Patel'
D. WHERE city = 'Pune'

Q281.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Isolation
C. Atomicity
D. Durability

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Q282.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Atomicity
C. Isolation
D. Consistency

Q283.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. Every row from A is retained; unmatched B columns are NULL
C. The result is always a Cartesian product
D. Every row from B is retained

Q284.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Consistency
C. Durability
D. Isolation

Q285.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A partial dependency on part of the candidate key
C. A multivalued dependency only
D. No dependency is violated

Q286.

A table contains three non-NULL values [68, 80, 32]. What is AVG(amount) for these rows?

A. 80
B. 180
C. 60.00
D. 63.33

Q287.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Durability
C. Isolation
D. Atomicity

Q288.

What does a CROSS JOIN produce?

A. Cartesian product of the input row sets
B. Only matching key rows
C. Only unmatched left rows
D. A union with duplicate elimination

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Q289.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE last_name = 'Patel'
C. WHERE city = 'Pune'
D. WHERE salary = 50000

Q290.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission == NULL
B. commission <> NULL
C. commission = NULL
D. commission IS NULL

Q291.

Consider salaries [51, 10, 18]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 61
B. 102
C. 79
D. 51

Q292.

Consider salaries [63, 87, 10]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 213
B. 150
C. 160
D. 237

Q293.

Consider salaries [81, 54, 89]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 224
B. 224
C. 305
D. 278

Q294.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. The result is always a Cartesian product
C. Every row from B is retained
D. Every row from A is retained; unmatched B columns are NULL

Q295.

For values [29, 30, 60], what is COUNT(*) returned by WHERE value > 30?

A. 3
B. 2
C. 2
D. 1

Q296.

For values [26, 19, 46], what is COUNT(*) returned by WHERE value > 19?

A. 2
B. 1
C. 3
D. 3

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Q297.

Consider salaries [66, 22, 28]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 116
B. 88
C. 132
D. 66

Q298.

A table contains three non-NULL values [88, 50, 81]. What is AVG(amount) for these rows?

A. 219
B. 88
C. 73.00
D. 76.33

Q299.

For values [66, 32, 77], what is COUNT(*) returned by WHERE value > 32?

A. 3
B. 1
C. 2
D. 3

Q300.

Consider salaries [57, 34, 37]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 57
B. 91
C. 128
D. 114

Q301.

For values [86, 46, 37], what is COUNT(*) returned by WHERE value > 46?

A. 3
B. 2
C. 2
D. 1

Q302.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Isolation
B. Consistency
C. Durability
D. Atomicity

Q303.

Which SQL clause sorts the final result set?

A. HAVING
B. ORDER BY
C. FROM
D. GROUP BY

Q304.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE last_name = 'Patel'
C. WHERE salary = 50000
D. WHERE first_name = 'Asha' only

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Q305.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. Only matching rows from A and B are retained
C. Every row from A is retained; unmatched B columns are NULL
D. The result is always a Cartesian product

Q306.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A multivalued dependency only
C. A transitive dependency through a non-key attribute
D. A partial dependency on part of the candidate key

Q307.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. No dependency is violated
C. A transitive dependency through a non-key attribute
D. A partial dependency on part of the candidate key

Q308.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission == NULL
C. commission IS NULL
D. commission <> NULL

Q309.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE salary = 50000
C. WHERE first_name = 'Asha' only
D. WHERE last_name = 'Patel'

Q310.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Atomicity
C. Durability
D. Isolation

Q311.

Consider salaries [34, 42, 47]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 34
B. 42
C. 123
D. 0

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Q312.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Consistency
C. Isolation
D. Atomicity

Q313.

Consider salaries [44, 63, 71]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 197
B. 178
C. 178
D. 134

Q314.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. The result is always a Cartesian product
C. Every row from A is retained; unmatched B columns are NULL
D. Every row from B is retained

Q315.

For values [25, 82, 46], what is COUNT(*) returned by WHERE value > 82?

A. 3
B. 1
C. 3
D. 0

Q316.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE salary = 50000
C. WHERE city = 'Pune'
D. WHERE last_name = 'Patel'

Q317.

A table contains three non-NULL values [10, 27, 56]. What is AVG(amount) for these rows?

A. 56
B. 34.33
C. 93
D. 31.00

Q318.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE city = 'Pune'
C. WHERE salary = 50000
D. WHERE last_name = 'Patel'

Q319.

For values [25, 66, 31], what is COUNT(*) returned by WHERE value > 66?

A. 3
B. 0
C. 3
D. 1

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Q320.

Which isolation anomaly involves the same query returning different sets of rows due to another committed transaction?

A. Dirty write only
B. Deadlock
C. Lost schema
D. Phantom read

Q321.

A table contains three non-NULL values [75, 26, 60]. What is AVG(amount) for these rows?

A. 53.67
B. 75
C. 161
D. 57.00

Q322.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. The result is always a Cartesian product
C. Every row from A is retained; unmatched B columns are NULL
D. Only matching rows from A and B are retained

Q323.

What does a database view generally represent?

A. A transaction log
B. A stored query definition presented as a virtual table
C. A unique index only
D. A physical copy of every source table

Q324.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE first_name = 'Asha' only
C. WHERE last_name = 'Patel'
D. WHERE salary = 50000

Q325.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission == NULL
C. commission IS NULL
D. commission = NULL

Q326.

Consider salaries [67, 13, 82]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 149
B. 216
C. 162
D. 162

Q327.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. No dependency is violated
C. A partial dependency on part of the candidate key
D. A transitive dependency through a non-key attribute

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Q328.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE first_name = 'Asha' only
C. WHERE last_name = 'Patel'
D. WHERE city = 'Pune'

Q329.

Consider salaries [15, 20, 13]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 15
B. 20
C. 48
D. 0

Q330.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE last_name = 'Patel'
B. WHERE salary = 50000
C. WHERE city = 'Pune'
D. WHERE first_name = 'Asha' only

Q331.

A table contains three non-NULL values [80, 51, 23]. What is AVG(amount) for these rows?

A. 54.67
B. 154
C. 51.33
D. 80

Q332.

Consider salaries [64, 34, 68]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 166
B. 166
C. 132
D. 196

Q333.

Consider salaries [85, 90, 55]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 230
B. 230
C. 315
D. 320

Q334.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from A is retained; unmatched B columns are NULL
B. Only matching rows from A and B are retained
C. The result is always a Cartesian product
D. Every row from B is retained

Q335.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Atomicity
C. Consistency
D. Isolation

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Q336.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE last_name = 'Patel'
C. WHERE city = 'Pune'
D. WHERE first_name = 'Asha' only

Q337.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. No dependency is violated
C. A multivalued dependency only
D. A partial dependency on part of the candidate key

Q338.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission == NULL
C. commission = NULL
D. commission <> NULL

Q339.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Atomicity
C. Isolation
D. Consistency

Q340.

Consider salaries [27, 86, 70]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 183
B. 156
C. 242
D. 183

Q341.

For values [21, 52, 59], what is COUNT(*) returned by WHERE value > 52?

A. 1
B. 3
C. 2
D. 2

Q342.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission = NULL
C. commission == NULL
D. commission <> NULL

Q343.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE salary = 50000
C. WHERE first_name = 'Asha' only
D. WHERE last_name = 'Patel'

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Q344.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A multivalued dependency only
C. A partial dependency on part of the candidate key
D. A transitive dependency through a non-key attribute

Q345.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Durability
C. Isolation
D. Consistency

Q346.

Consider salaries [49, 54, 24]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 54
B. 103
C. 127
D. 108

Q347.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A partial dependency on part of the candidate key
C. A transitive dependency through a non-key attribute
D. A multivalued dependency only

Q348.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from A is retained; unmatched B columns are NULL
B. Every row from B is retained
C. The result is always a Cartesian product
D. Only matching rows from A and B are retained

Q349.

A table contains three non-NULL values [52, 83, 42]. What is AVG(amount) for these rows?

A. 177
B. 62.33
C. 59.00
D. 83

Q350.

Consider salaries [54, 55, 87]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 196
B. 251
C. 196
D. 250

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Q351.

Consider salaries [19, 85, 39]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 104
B. 143
C. 85
D. 170

Q352.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Durability
C. Isolation
D. Consistency

Q353.

A table contains three non-NULL values [72, 74, 66]. What is AVG(amount) for these rows?

A. 74
B. 70.67
C. 212
D. 74.00

Q354.

A table contains three non-NULL values [56, 55, 31]. What is AVG(amount) for these rows?

A. 142
B. 47.33
C. 56
D. 50.67

Q355.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE city = 'Pune'
C. WHERE salary = 50000
D. WHERE last_name = 'Patel'

Q356.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Isolation
C. Consistency
D. Durability

Q357.

Consider salaries [15, 13, 39]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 0
B. 15
C. 13
D. 67

Q358.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE last_name = 'Patel'
C. WHERE city = 'Pune'
D. WHERE salary = 50000

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Q359.

What does a composite index cover?

A. All possible query plans
B. Only text columns
C. Only one row
D. Multiple fields in a defined key order

Q360.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A partial dependency on part of the candidate key
B. No dependency is violated
C. A multivalued dependency only
D. A transitive dependency through a non-key attribute

Q361.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. A partial dependency on part of the candidate key
C. A transitive dependency through a non-key attribute
D. No dependency is violated

Q362.

A table contains three non-NULL values [39, 22, 48]. What is AVG(amount) for these rows?

A. 48
B. 109
C. 39.67
D. 36.33

Q363.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. Only matching rows from A and B are retained
C. The result is always a Cartesian product
D. Every row from A is retained; unmatched B columns are NULL

Q364.

A table contains three non-NULL values [76, 75, 11]. What is AVG(amount) for these rows?

A. 54.00
B. 76
C. 162
D. 57.33

Q365.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission IS NULL
C. commission <> NULL
D. commission == NULL

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Q366.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Consistency
C. Durability
D. Isolation

Q367.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Consistency
C. Isolation
D. Atomicity

Q368.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE city = 'Pune'
C. WHERE last_name = 'Patel'
D. WHERE salary = 50000

Q369.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Isolation
C. Atomicity
D. Consistency

Q370.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission == NULL
C. commission = NULL
D. commission IS NULL

Q371.

Consider salaries [18, 38, 41]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 0
B. 18
C. 38
D. 97

Q372.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE city = 'Pune'
C. WHERE salary = 50000
D. WHERE last_name = 'Patel'

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Q373.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. No dependency is violated
C. A partial dependency on part of the candidate key
D. A transitive dependency through a non-key attribute

Q374.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. No dependency is violated
C. A multivalued dependency only
D. A partial dependency on part of the candidate key

Q375.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Consistency
C. Atomicity
D. Isolation

Q376.

For values [73, 70, 26], what is COUNT(*) returned by WHERE value > 70?

A. 3
B. 1
C. 2
D. 2

Q377.

For values [57, 56, 18], what is COUNT(*) returned by WHERE value > 56?

A. 1
B. 2
C. 2
D. 3

Q378.

Which command is generally classified as DDL?

A. COMMIT
B. INSERT
C. CREATE TABLE
D. UPDATE

Q379.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission IS NULL
C. commission == NULL
D. commission = NULL

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Q380.

What is the purpose of EXPLAIN in SQL systems?

A. Inspect the query execution plan
B. Create a trigger
C. Export a table
D. Commit a transaction

Q381.

Consider salaries [47, 45, 84]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 129
B. 131
C. 176
D. 84

Q382.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Isolation
B. Durability
C. Atomicity
D. Consistency

Q383.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A partial dependency on part of the candidate key
C. No dependency is violated
D. A multivalued dependency only

Q384.

A table contains three non-NULL values [84, 32, 75]. What is AVG(amount) for these rows?

A. 191
B. 63.67
C. 84
D. 67.00

Q385.

Which SQL construct returns a single value for each group?

A. ORDER BY alone
B. Aggregate function with GROUP BY
C. CREATE INDEX
D. DISTINCT alone

Q386.

Consider salaries [10, 54, 62]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 126
B. 116
C. 170
D. 126

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Q387.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Consistency
C. Durability
D. Isolation

Q388.

A table contains three non-NULL values [37, 21, 85]. What is AVG(amount) for these rows?

A. 143
B. 47.67
C. 85
D. 51.00

Q389.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. Every row from A is retained; unmatched B columns are NULL
C. The result is always a Cartesian product
D. Every row from B is retained

Q390.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission == NULL
C. commission = NULL
D. commission <> NULL

Q391.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. No dependency is violated
C. A multivalued dependency only
D. A partial dependency on part of the candidate key

Q392.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. Every row from B is retained
C. The result is always a Cartesian product
D. Every row from A is retained; unmatched B columns are NULL

Q393.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE salary = 50000
C. WHERE city = 'Pune'
D. WHERE last_name = 'Patel'

Q394.

Consider salaries [48, 71, 63]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 182
B. 205
C. 134
D. 182

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Q395.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission == NULL
C. commission <> NULL
D. commission IS NULL

Q396.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Isolation
B. Atomicity
C. Consistency
D. Durability

Q397.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE last_name = 'Patel'
B. WHERE salary = 50000
C. WHERE city = 'Pune'
D. WHERE first_name = 'Asha' only

Q398.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. Every row from B is retained
C. Every row from A is retained; unmatched B columns are NULL
D. The result is always a Cartesian product

Q399.

For values [36, 85, 21], what is COUNT(*) returned by WHERE value > 85?

A. 3
B. 3
C. 1
D. 0

Q400.

Consider salaries [41, 51, 37]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 92
B. 102
C. 51
D. 129

Q401.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE last_name = 'Patel'
B. WHERE salary = 50000
C. WHERE first_name = 'Asha' only
D. WHERE city = 'Pune'

Q402.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission == NULL
C. commission IS NULL
D. commission <> NULL

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Q403.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE last_name = 'Patel'
B. WHERE salary = 50000
C. WHERE first_name = 'Asha' only
D. WHERE city = 'Pune'

Q404.

A table contains three non-NULL values [43, 83, 68]. What is AVG(amount) for these rows?

A. 194
B. 64.67
C. 83
D. 68.00

Q405.

Consider salaries [14, 11, 77]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 91
B. 88
C. 77
D. 102

Q406.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Only matching rows from A and B are retained
C. Every row from A is retained; unmatched B columns are NULL
D. Every row from B is retained

Q407.

Consider salaries [81, 48, 40]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 81
B. 162
C. 169
D. 129

Q408.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Isolation
B. Durability
C. Consistency
D. Atomicity

Q409.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE salary = 50000
C. WHERE city = 'Pune'
D. WHERE last_name = 'Patel'

Q410.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission == NULL
B. commission <> NULL
C. commission IS NULL
D. commission = NULL

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Q411.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE salary = 50000
C. WHERE first_name = 'Asha' only
D. WHERE last_name = 'Patel'

Q412.

Which MongoDB operation modifies matching documents?

A. updateMany()
B. countDocuments()
C. find()
D. aggregate() only

Q413.

What is a correlated subquery?

A. A query that must return multiple rows
B. A subquery that executes only once in every DBMS
C. A subquery that references columns from the outer query
D. A query with no WHERE clause

Q414.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. Every row from A is retained; unmatched B columns are NULL
C. Every row from B is retained
D. The result is always a Cartesian product

Q415.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission IS NULL
C. commission == NULL
D. commission <> NULL

Q416.

Consider salaries [49, 26, 55]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 104
B. 81
C. 55
D. 130

Q417.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Durability
C. Isolation
D. Atomicity

Q418.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A partial dependency on part of the candidate key
C. No dependency is violated
D. A multivalued dependency only

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Q419.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from B is retained
C. Every row from A is retained; unmatched B columns are NULL
D. Only matching rows from A and B are retained

Q420.

For values [23, 81, 51], what is COUNT(*) returned by WHERE value > 81?

A. 3
B. 3
C. 0
D. 1

Q421.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. A transitive dependency through a non-key attribute
C. A partial dependency on part of the candidate key
D. No dependency is violated

Q422.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A multivalued dependency only
C. A transitive dependency through a non-key attribute
D. A partial dependency on part of the candidate key

Q423.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from B is retained
C. Every row from A is retained; unmatched B columns are NULL
D. Only matching rows from A and B are retained

Q424.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Durability
C. Atomicity
D. Isolation

Q425.

A table contains three non-NULL values [70, 88, 75]. What is AVG(amount) for these rows?

A. 81.00
B. 233
C. 77.67
D. 88

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Q426.

For values [14, 62, 57], what is COUNT(*) returned by WHERE value > 62?

A. 3
B. 1
C. 0
D. 3

Q427.

Consider salaries [70, 34, 14]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 118
B. 104
C. 140
D. 70

Q428.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. Every row from B is retained
C. Every row from A is retained; unmatched B columns are NULL
D. The result is always a Cartesian product

Q429.

Which property ensures committed data survives a system crash, subject to the DBMS durability guarantees?

A. Consistency
B. Isolation
C. Durability
D. Atomicity

Q430.

Consider salaries [75, 48, 39]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 75
B. 123
C. 150
D. 162

Q431.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission = NULL
C. commission <> NULL
D. commission == NULL

Q432.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE city = 'Pune'
B. WHERE salary = 50000
C. WHERE last_name = 'Patel'
D. WHERE first_name = 'Asha' only

Q433.

A table contains three non-NULL values [32, 36, 34]. What is AVG(amount) for these rows?

A. 34.00
B. 102
C. 36
D. 37.33

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Q434.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE city = 'Pune'
C. WHERE last_name = 'Patel'
D. WHERE first_name = 'Asha' only

Q435.

What does UNION do by default?

A. Combines columns horizontally
B. Combines result sets and removes duplicate rows
C. Keeps all duplicates
D. Updates matching records

Q436.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. No dependency is violated
C. A multivalued dependency only
D. A partial dependency on part of the candidate key

Q437.

Consider salaries [58, 43, 90]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 191
B. 191
C. 148
D. 206

Q438.

A table contains three non-NULL values [42, 21, 55]. What is AVG(amount) for these rows?

A. 55
B. 118
C. 39.33
D. 42.67

Q439.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. The result is always a Cartesian product
B. Every row from A is retained; unmatched B columns are NULL
C. Only matching rows from A and B are retained
D. Every row from B is retained

Q440.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Isolation
B. Consistency
C. Durability
D. Atomicity

Q441.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE last_name = 'Patel'
B. WHERE salary = 50000
C. WHERE first_name = 'Asha' only
D. WHERE city = 'Pune'

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Q442.

Consider salaries [40, 13, 74]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 114
B. 87
C. 127
D. 74

Q443.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission <> NULL
C. commission == NULL
D. commission = NULL

Q444.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. A partial dependency on part of the candidate key
C. A transitive dependency through a non-key attribute
D. No dependency is violated

Q445.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. A partial dependency on part of the candidate key
C. No dependency is violated
D. A transitive dependency through a non-key attribute

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Q446.

For values [68, 89, 86], what is COUNT(*) returned by WHERE value > 89?

A. 3
B. 3
C. 0
D. 1

Q447.

For values [66, 10, 23], what is COUNT(*) returned by WHERE value > 10?

A. 3
B. 1
C. 2
D. 3

Q448.

A table contains three non-NULL values [37, 17, 21]. What is AVG(amount) for these rows?

A. 28.33
B. 37
C. 75
D. 25.00

Q449.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission == NULL
C. commission IS NULL
D. commission <> NULL

Q450.

A table contains three non-NULL values [59, 35, 86]. What is AVG(amount) for these rows?

A. 180
B. 86
C. 63.33
D. 60.00

Q451.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. The result is always a Cartesian product
C. Every row from B is retained
D. Every row from A is retained; unmatched B columns are NULL

Q452.

A table contains three non-NULL values [15, 47, 20]. What is AVG(amount) for these rows?

A. 27.33
B. 82
C. 47
D. 30.67

Q453.

A table contains three non-NULL values [14, 50, 34]. What is AVG(amount) for these rows?

A. 32.67
B. 36.00
C. 50
D. 98

Q454.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A transitive dependency through a non-key attribute
C. A multivalued dependency only
D. A partial dependency on part of the candidate key

Q455.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A transitive dependency through a non-key attribute
C. A partial dependency on part of the candidate key
D. A multivalued dependency only

Q456.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Durability
C. Isolation
D. Atomicity

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Q457.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Isolation
B. Consistency
C. Durability
D. Atomicity

Q458.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Isolation
C. Durability
D. Consistency

Q459.

Which SQL clause filters groups after aggregation?

A. HAVING
B. WHERE
C. ORDER BY
D. LIMIT

Q460.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Isolation
B. Consistency
C. Atomicity
D. Durability

Q461.

For values [54, 37, 40], what is COUNT(*) returned by WHERE value > 37?

A. 2
B. 3
C. 3
D. 1

Q462.

For values [20, 68, 68], what is COUNT(*) returned by WHERE value > 68?

A. 1
B. 3
C. 3
D. 0

Q463.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE first_name = 'Asha' only
C. WHERE last_name = 'Patel'
D. WHERE city = 'Pune'

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Q464.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Durability
C. Consistency
D. Isolation

Q465.

For values [25, 44, 45], what is COUNT(*) returned by WHERE value > 44?

A. 3
B. 2
C. 2
D. 1

Q466.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Atomicity
B. Isolation
C. Consistency
D. Durability

Q467.

Consider salaries [83, 45, 34]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 128
B. 166
C. 83
D. 162

Q468.

Consider salaries [25, 40, 78]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 78
B. 118
C. 143
D. 103

Q469.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Only matching rows from A and B are retained
B. Every row from A is retained; unmatched B columns are NULL
C. Every row from B is retained
D. The result is always a Cartesian product

Q470.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission == NULL
B. commission IS NULL
C. commission <> NULL
D. commission = NULL

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Q471.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. A transitive dependency through a non-key attribute
C. No dependency is violated
D. A partial dependency on part of the candidate key

Q472.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission IS NULL
B. commission = NULL
C. commission <> NULL
D. commission == NULL

Q473.

A table contains three non-NULL values [73, 66, 67]. What is AVG(amount) for these rows?

A. 206
B. 68.67
C. 73
D. 72.00

Q474.

A table contains three non-NULL values [69, 69, 55]. What is AVG(amount) for these rows?

A. 67.67
B. 64.33
C. 69
D. 193

Q475.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A multivalued dependency only
B. A partial dependency on part of the candidate key
C. No dependency is violated
D. A transitive dependency through a non-key attribute

Q476.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A partial dependency on part of the candidate key
C. A transitive dependency through a non-key attribute
D. A multivalued dependency only

Q477.

For values [39, 78, 89], what is COUNT(*) returned by WHERE value > 78?

A. 2
B. 1
C. 2
D. 3

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Q478.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission = NULL
B. commission IS NULL
C. commission <> NULL
D. commission == NULL

Q479.

A table contains three non-NULL values [33, 46, 82]. What is AVG(amount) for these rows?

A. 82
B. 57.00
C. 161
D. 53.67

Q480.

A table contains three non-NULL values [49, 85, 15]. What is AVG(amount) for these rows?

A. 85
B. 53.00
C. 49.67
D. 149

Q481.

For values [32, 35, 12], what is COUNT(*) returned by WHERE value > 35?

A. 3
B. 1
C. 0
D. 3

Q482.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Durability
B. Atomicity
C. Consistency
D. Isolation

Q483.

For values [16, 73, 63], what is COUNT(*) returned by WHERE value > 73?

A. 0
B. 1
C. 3
D. 3

Q484.

A transaction updates two account rows and the system fails between the updates before commit. Which ACID property requires the partial work to be undone?

A. Consistency
B. Atomicity
C. Isolation
D. Durability

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Q485.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. A transitive dependency through a non-key attribute
B. A multivalued dependency only
C. A partial dependency on part of the candidate key
D. No dependency is violated

Q486.

Which key uniquely identifies each row and cannot contain NULL?

A. Foreign key
B. Partial index
C. Primary key
D. Alternate non-unique key

Q487.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE first_name = 'Asha' only
B. WHERE city = 'Pune'
C. WHERE last_name = 'Patel'
D. WHERE salary = 50000

Q488.

A table contains three non-NULL values [24, 27, 39]. What is AVG(amount) for these rows?

A. 39
B. 30.00
C. 90
D. 33.33

Q489.

For values [61, 90, 26], what is COUNT(*) returned by WHERE value > 90?

A. 3
B. 1
C. 3
D. 0

Q490.

What is the purpose of $project in a MongoDB aggregation pipeline?

A. Filter documents only
B. Sort documents only
C. Reshape or select fields in pipeline documents
D. Join collections only

Q491.

A table contains three non-NULL values [57, 53, 59]. What is AVG(amount) for these rows?

A. 169
B. 56.33
C. 59.67
D. 59

Q492.

For values [14, 61, 20], what is COUNT(*) returned by WHERE value > 61?

A. 1
B. 3
C. 0
D. 3

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Q493.

Which MongoDB operator is commonly used to match documents against a condition?

A. $match
B. $group
C. $sort
D. $project

Q494.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE last_name = 'Patel'
B. WHERE first_name = 'Asha' only
C. WHERE salary = 50000
D. WHERE city = 'Pune'

Q495.

A table has a composite primary key (student_id, course_id). Which dependency violates 2NF when student_name depends only on student_id?

A. No dependency is violated
B. A transitive dependency through a non-key attribute
C. A partial dependency on part of the candidate key
D. A multivalued dependency only

Q496.

Which predicate correctly selects rows where commission is unknown (NULL)?

A. commission <> NULL
B. commission == NULL
C. commission IS NULL
D. commission = NULL

Q497.

A B-tree index exists on (last_name, first_name). Which predicate is most directly supported by the leftmost-prefix rule?

A. WHERE salary = 50000
B. WHERE first_name = 'Asha' only
C. WHERE city = 'Pune'
D. WHERE last_name = 'Patel'

Q498.

Which join returns only rows with matching join keys in both inputs?

A. LEFT JOIN
B. FULL OUTER JOIN
C. CROSS JOIN
D. INNER JOIN

Q499.

A query uses LEFT JOIN from table A to table B. Which statement is correct?

A. Every row from B is retained
B. The result is always a Cartesian product
C. Every row from A is retained; unmatched B columns are NULL
D. Only matching rows from A and B are retained

Q500.

Consider salaries [36, 81, 69]. What does SELECT SUM(salary) FROM Employee WHERE salary >= 50 return?

A. 186
B. 186
C. 231
D. 150

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CONSOLIDATED ANSWER KEY

Answers are listed by question number. Each key corresponds to the option order shown in the question section.

1. C	2. A	3. C	4. B	5. C
6. A	7. D	8. A	9. C	10. C
11. D	12. A	13. B	14. D	15. D
16. D	17. A	18. D	19. C	20. C
21. B	22. A	23. B	24. B	25. A
26. B	27. D	28. C	29. C	30. C
31. A	32. B	33. B	34. A	35. A
36. C	37. C	38. B	39. A	40. C
41. D	42. D	43. A	44. D	45. A
46. A	47. B	48. A	49. C	50. D
51. C	52. A	53. D	54. C	55. C
56. D	57. A	58. C	59. C	60. A
61. A	62. D	63. B	64. B	65. D
66. B	67. C	68. A	69. D	70. D
71. C	72. B	73. A	74. B	75. C
76. B	77. C	78. B	79. C	80. B
81. A	82. B	83. B	84. C	85. C
86. C	87. C	88. A	89. B	90. D
91. C	92. B	93. C	94. C	95. A
96. B	97. A	98. D	99. D	100. D
101. C	102. C	103. C	104. A	105. C
106. D	107. A	108. A	109. C	110. C
111. A	112. D	113. D	114. C	115. C
116. B	117. C	118. A	119. C	120. B
121. B	122. A	123. D	124. D	125. C
126. B	127. A	128. D	129. A	130. B
131. A	132. C	133. A	134. D	135. D
136. B	137. B	138. C	139. D	140. D
141. D	142. D	143. B	144. B	145. A
146. C	147. A	148. B	149. A	150. C
151. A	152. B	153. B	154. B	155. C

156. A	157. C	158. C	159. D	160. B
161. D	162. B	163. D	164. D	165. A
166. A	167. C	168. A	169. B	170. B
171. C	172. A	173. D	174. C	175. A
176. C	177. C	178. C	179. C	180. B
181. D	182. D	183. B	184. C	185. D
186. C	187. B	188. B	189. D	190. B
191. C	192. B	193. A	194. C	195. D
196. C	197. A	198. D	199. D	200. C
201. B	202. D	203. C	204. C	205. A
206. B	207. A	208. A	209. B	210. C
211. A	212. B	213. A	214. D	215. C
216. B	217. C	218. A	219. C	220. A
221. B	222. D	223. C	224. B	225. B
226. B	227. C	228. B	229. C	230. C
231. D	232. D	233. A	234. D	235. C
236. B	237. B	238. A	239. D	240. A
241. B	242. D	243. B	244. B	245. A
246. C	247. A	248. A	249. B	250. C
251. A	252. B	253. B	254. C	255. B
256. A	257. B	258. B	259. D	260. B
261. B	262. C	263. A	264. D	265. C
266. C	267. B	268. C	269. C	270. C
271. B	272. B	273. C	274. A	275. C
276. A	277. A	278. A	279. B	280. C
281. C	282. B	283. B	284. A	285. B
286. C	287. D	288. A	289. B	290. D
291. D	292. B	293. B	294. D	295. D
296. A	297. D	298. C	299. C	300. A
301. D	302. D	303. B	304. B	305. C
306. D	307. D	308. C	309. D	310. B
311. D	312. D	313. D	314. C	315. D
316. D	317. D	318. D	319. B	320. D

321. A	322. C	323. B	324. C	325. C
326. A	327. C	328. C	329. D	330. A
331. C	332. C	333. B	334. A	335. B
336. B	337. D	338. A	339. B	340. B
341. A	342. A	343. D	344. C	345. A
346. A	347. B	348. A	349. C	350. A
351. C	352. A	353. B	354. B	355. D
356. A	357. A	358. B	359. D	360. A
361. B	362. D	363. D	364. A	365. B
366. A	367. D	368. C	369. C	370. D
371. A	372. D	373. C	374. D	375. C
376. B	377. A	378. C	379. B	380. A
381. D	382. C	383. B	384. B	385. B
386. B	387. A	388. B	389. B	390. A
391. D	392. D	393. D	394. C	395. D
396. B	397. A	398. C	399. D	400. C
401. A	402. C	403. A	404. B	405. C
406. C	407. A	408. D	409. D	410. C
411. D	412. A	413. C	414. B	415. B
416. C	417. D	418. B	419. C	420. C
421. C	422. D	423. C	424. C	425. C
426. C	427. D	428. C	429. C	430. A
431. A	432. C	433. A	434. C	435. B
436. D	437. C	438. C	439. B	440. D
441. A	442. D	443. A	444. B	445. B
446. C	447. C	448. D	449. C	450. D
451. D	452. A	453. A	454. D	455. C
456. D	457. D	458. A	459. A	460. C
461. A	462. D	463. C	464. A	465. D
466. A	467. C	468. A	469. B	470. B
471. D	472. A	473. B	474. B	475. B
476. B	477. B	478. B	479. D	480. C
481. C	482. B	483. A	484. B	485. C

486. C	**487. C**	**488. B**	**489. D**	**490. C**
491. B	**492. C**	**493. A**	**494. A**	**495. C**
496. C	**497. D**	**498. D**	**499. C**	**500. D**

Reference basis: C-DAC official PG-DAC/PGCP-AC course brochure and publicly available module outlines for topic coverage; publicly available C-DAC evaluation guidance for the objective-question format and difficulty framework. Questions in this booklet are independently authored practice items, not copied past-paper questions.

Official course outline: https://cdac.gov.in/index.aspx?courseid=0&id=DAC
C-DAC evaluation guideline reference: publicly available C-DAC evaluation guideline PDF (search result/source reference used during preparation).

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